Indian Institute of Information Technology Vadodara

CS/IT 429

Hardware for Embedded System Design

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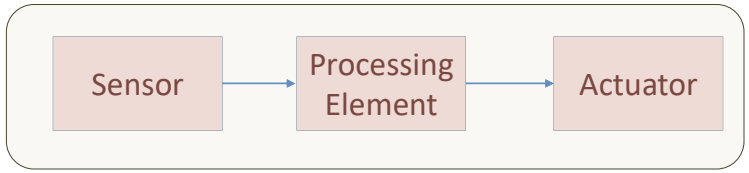
Hardware for Embedded System Design

- A Processor is the heart of the Embedded System.
- Embedded systems widely incorporates micro-controller units (MCUs).
- Historical microcontrollers were specialized circuits (such as Programmable Logic Device, Programmable ROM) made with array of logic gates.
- However, newer microcontrollers contain a low-complexity microprocessor core along with some memories and peripherals commonly needed for embedded system designs.



History of Microcontrollers

- History of microcontroller goes back to first microcontrollers that originated from digital logics. First microcontroller is credited to Texas Instruments (TI) engineers Gary Boone and Michael Cochran who developed TMS1000 in 1971. Early microcontrollers were logic array-based architecture and aimed to integrate multiple digital logics on a single chip.
- Later Intel developed 4004 and 8048 (microprocessors) that laid the foundation for newer microcontrollers. Instead of a sea of logic arrays, newer microcontroller uses microarchitecture (architecture of a microprocessor) that allows persistent software code (firmware) to run on the microcontroller hardware, instead of only hardware-based execution in the logic arrays.

Microprocessor vs. Microcontroller

Microprocessor	Microcontroller
A microprocessor is the main component of a computer.	A microcontroller is itself a small computer system-on-a-chip.
The functional blocks are ALU, registers, timing and control units.	It contains functional blocks of microprocessors and in addition has timer, parallel i/o, memory such as RAM, EPROM, ADC and DAC.
It has high processing capabilities.	It has low processing capabilities.
It consumes high power.	It consumes low power.
It is generally used for general purpose computing (e.g. personal computer, server)	It is generally used in embedded systems (e.g. car, modem, TV, IoT, wearables)
Some Examples: Intel: 8086, 8088, 80186, 80188, 80286, 80386 Motorola: 6800, 6809, 68000, G3, G4, G5	Some Examples: Intel: 8032, 8051, 8052 PIC: 8-bit PIC16, PIC18 16-bit DSPIC33/PIC24 Motorola: 6811

Microprocessor vs. Microcontroller

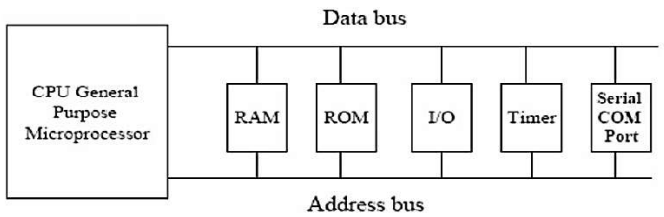
- Microcontrollers contain one or more CPUs with RAM, ROM and programmable input/output peripherals.

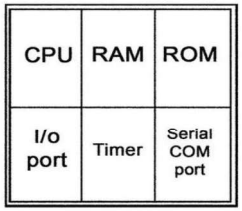
- Microprocessors tend to operate at much greater clock speeds on general application tasks, like gaming, photo editing and software development, etc.
  whereas microcontrollers are designed for more specific tasks in smaller embedded systems like keyboards, mice, electronic toys and vending machines.

- In short, it can be easier to turn on and off an LED with a microcontroller as it has direct access to the output pin, whereas microprocessors have to communicate with the external peripherals (ICs) to control the LED.

Microprocessor vs. Microcontroller

A single chip Microcontroller





(Single chip)

Selecting the right Microcontroller for an ES…!

Key Features

- Application

  The application dictates which type of microcontroller and the features.
  Some common applications include wearables, automotive, industrial, smart home/energy and the internet of things.

- Key points of consideration

  Will the application need a lot of processing?

  Can this processing be done in an 8bit microcontroller? How complex is the design? If the design needs to perform numerous calculations, more memory may be needed.

Example

- Performing an FFT on an audio signal will require more RAM, greater sampling rates and high-resolution ADCs, as well as other features.
- Taking a temperature reading and sending commands over UART are relatively light weight and much smaller microcontrollers can be used with less memory.

[https://octopart.com]

Selecting the right Microcontroller for an ES…!

Key Features

- Bit Size

  The bit size actually refers to the size of a 'word' used inside the processor. This 'word' is a piece of data handled by the instruction set of the microcontroller.
  The size of the word differs in each microcontroller between 8 bits, 16 bits, 32 bits and even 64 bits.
  A word can be used as an address, a variable, a register or an instruction. In an 8-bit processor, 8 bits are used in every instruction. 16-bit and 32-bit processors can operate with larger amounts of data. Larger register or bus widths, means less restrictions on resources.

This also has an effect on the size of internal ROM and RAM memory. In the 8-bit processor again, the memory size is dictated by 8 bits, which gives 255 unique memory locations. Whereas a 16-bit microcontroller can handle up to 65,535 memory locations and a 32-bit microcontroller can handle up to 4,294,967,295 memory locations.

Example

- An 8-bit microcontroller will be sufficient when communicating with low speed buses, taking sensor measurements or even controlling a buzzer. However, it will have a lot of trouble trying to control an LCD touch screen or Ethernet interface. This is where a 32-bit microcontroller will excel.

[https://octopart.com]

Selecting the right Microcontroller for an ES…!

Key Features

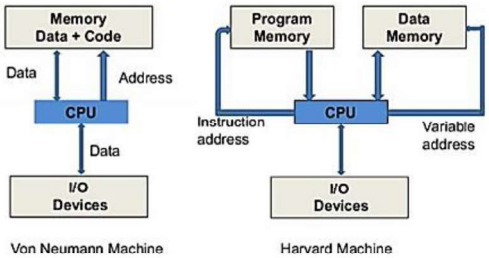
- Architecture

  There are different architectural approaches which are incorporated in varying processors and controllers.

Basics of Computer

Types of Architecture

- Von Neumann Architecture [Princeton Architecture]
- Harvard Architecture



Basics of Computer

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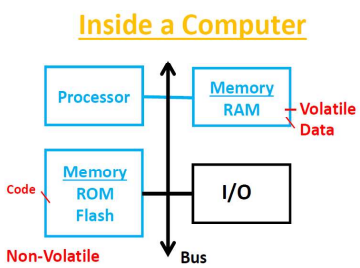
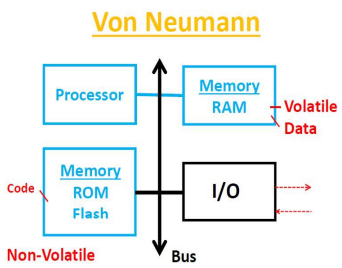
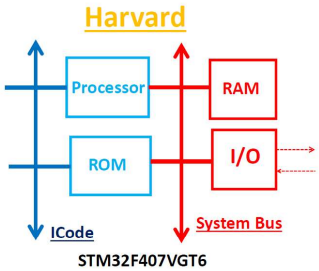






Basics of Computer

Instruction Set Architecture (ISA)

- An instruction is made up of two things, its opcode (what the instruction is asking the computer to do) and its operands (who the computer is expected to do it to).
- The instruction set is defined by its set of opcodes, data types, and addressing modes. The addressing modes determine where the operands are located. The data type is the representation of the operands in 0s and 1s.
- The main ISAs are RISC and CISC:
  RISC stands for Reduced Instruction Set Architecture
  CISC stands for Complex Instruction Set Architecture

RISC and CISC Instructions

- RISC – Reduced Instruction Set Architecture

  - The main idea behind this is to make hardware simpler by using an instruction set composed of a few basic steps for loading, evaluating, and storing operations just like a load command will load data, a store command will store the data. It is also called Load-Store Architecture.

  - Both RISC and CISC approaches try to increase the CPU Performance.

  RISC approach reduces the cycles per instructions at the cost of the number of instructions per program.

- CISC – Complex Instruction Set Architecture

  - The main idea is that a single instruction will do all loading, evaluating, and storing operations just like a multiplication command with do stuff like loading data, multiplication, and storing it, hence its complex.

  - The CISC approach attempts to minimize the number of instructions per program but at the cost of an increase in the number of cycles per instruction.

RISC and CISC Instructions

Both RISC and CISC approaches try to increase the CPU Performance.

- RISC – Reduced Instruction Set Architecture

  RISC approach reduces the cycles per instructions at the cost of the number of instructions per program.

- CISC – Complex Instruction Set Architecture

  The CISC approach attempts to minimize the number of instructions per program but at the cost of an increase in the number of cycles per instruction.

$$CPU\ Time = \frac{Seconds}{Program} = \frac{Instructions}{Program} X \frac{Cycles}{Instructions} X \frac{Seconds}{Cycle}$$

Key Features

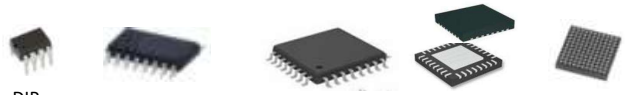
- Package Size

  ICs are the backbone of electronic devices, and choosing the right IC requires considering factors like price, reliability, performance, and package, among others.

  There are many different types of IC packages which differ by size, pin count, pin location and mounting type — surface-mount or through-hole.

Example

- To start off on a breadboard, then a Dual Inline Package (DIP) package will be necessary. This isn't usually available for many of the microcontrollers. Alternatively, Small Outline Integrated Circuit (SOIC), shrink small outline package (SSOP) and thin-shrink small outline package (TSSOP) parts are relatively easy enough to solder by hand, where Quad Flat No-lead (QFN), Quad Flat Package (QFP) and Ball Grid Array (BGA)/Land Grid Array (LGA) devices need proper reflow to assemble.

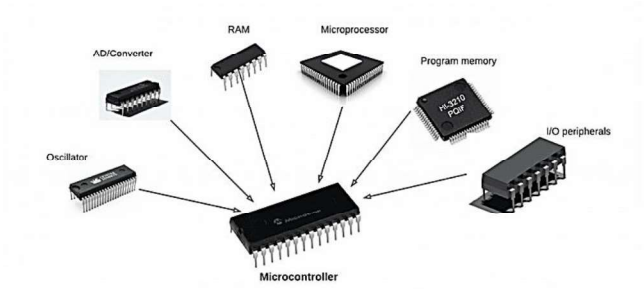


DIP SOIC QFP QFN BGA

[https://octopart.com]

Key Features

- Package Size



A microcontroller system is embedded in an integrated circuit (IC). A typical microcontroller includes a processor, program memory, RAM, input/output pins, and more on a single chip.

[https://octopart.com]

Key Features

- Communication Interfaces

Example

- A serial EEPROM may need a I2C or SPI interface. Other devices may require UARTs/USART/EUSARTs for serial communication. Some displays require parallel interfaces, which can be pin heavy.

[https://octopart.com]

Key Features

- Memory

- There are different memory types available in each microcontroller.
- These are usually volatile memory; RAM and ROM.
- There are a few microcontrollers that have non-volatile memory EEPROMs as well.
- It can be difficult to gauge how much of each is needed at first and takes some experience to get the hang of it.
- This is also dictated by the bit size mentioned earlier.
- It's important to note the difference between program and data memory.
- Program memory is where the compiled code will be stored and run from and the data memory is used during run time. If a lot of processing is needed, then more data memory will be needed. If the project needs to use look up tables or store a lot of pre-defined values, then larger program memory will be needed.

[https://octopart.com]

Various Microcontrollers

|*Vahid/Givargis, Embedded Systems Design: A Unified Hardware/Software Introduction*|

- PLD, PLA and PAL
- FPLA and FPGA
- ASIC
- Microcontroller Unit

PLD, PLA and PAL

- The earlier microcontrollers are based on logic arrays or programmable read-only memory (PROM) that contained an array of wire-interconnects that were programmable (burnt for shorts) only once.
- Based on the program, certain inputs would produce certain outputs. These were known as Programmable Logic Device (PLD), Programmable Logic Array (PLA), or Programmable Array Logic (PAL).
- Connections on the chip were either created or destroyed to implement the desired functionality.
- Advantage: These devices were reduced NRE costs and almost instantiations availability of programmed microcontrollers.
  Disadvantage: But they were power hungry, slow, and had a low processing power.

FPLA and FPGA

- Field Programmable Logic Array (FPLA) was introduced around 1975 by Signetics. It aimed to replace and improve upon multiple logic chips (e.g., AND, OR, NOT, NAND, Multiplexor, etc.) that were typically used at that time to develop embedded digital controllers.
- FPLA integrated many logic gates in a single chip and allowed the users to be able to program the interconnects among these logic gates based on their needs. An FPLA chip would contain multiple input logic lines that feed to a series of AND (input gates), followed by OR and XOR gates (summing matrix). The interconnects were made with programmable links.
- Later, in the 1980s, a newer technology, named Field Programmable Gate Array (FPGA) came to market. There were two main manufacturers of FPGA: Altera (founded in 1983) and Xilinx (founded in 1985). These companies advanced FPLA to a new generation of FPGA. These devices integrated basic logic gate combinations in units called Configurable Logic Blocks (CLB). These logic blocks are designed and optimized to provide reconfigurable logic gates. Programmable logic blocks were invented by David W. Page and LuVerne R. Peterson, and patented in 1985.
  The CLBs typically consist of some Look-up tables (LUTs), followed by combinational logic blocks like Full Adders, and sequential logic blocks like D-Flip-Flop. The CLBs are connected with programmable routing wires (i.e., switch box). Most FPGAs also have memories (e.g., RAM block), Delay Locked Loop (DLL), times, and input/output (I/O) cells. The users can design custom logics with Hardware Description Language (HDL) code that can be synthesized to be downloaded in the FPGA to perform control algorithms.
- The traditional FPGAs would only allow hardware-based microcontroller designs. But newer FPGAs contain built-in microprocessors in addition to reconfigurable hardware. For instance, newer Altera FPGAs (e.g., Cyclone series) contain IP Cores (viz. Soft Microcontroller) named Nios II. This expands the capabilities of FPGAs to perform signal processing with both hardware and software as needed. Due to fast speed (of hardware), FPGAs are suitable for high-speed processing and Finite State Machines (FSM). But they are typically expensive compared to other simpler microcontrollers and complex, hence the design requirement must warrant use of FPGA for an embedded system application.

ASIC

- Microcontrollers could be designed with custom IC fabrication process such as Application Specific Integrated Circuit (ASIC). An ASIC is a full-custom IC device such as a microcontroller that is designed for specific applications. A fullcustom IC results in the minimal waste of silicon chip area. However, designing ASIC chip is a highly complex, time consuming, and expensive process. Thus, this approach leads to a huge NRE, thus not suitable for most embedded system applications. Only exceptions are when the volume of production is large enough to compensate for the NRE, or the application requires high optimality while the expense is irrelevant. Examples may include implantable pacemakers, brain implants, and specific-purpose and delicate bioelectric signal acquisition chip.
- Advantage: Full-custom ASIC allows all layers of chip design to be optimized by the designers, which includes placement of transistors, sizing of transistors, and routing of interconnect wires. This leads to even optimize performance, smallest size, and lowest power consumption.
  Disadvantage: High NRE cost (often more than $500 k) and long time-to-market often offsets the advantages. Semicustom ASIC are faster to design and develop, hence leads to lower NRE. In this case, lower layers of chip design are fully or partially built. Chip designers only customize the routing wires and maybe place some additional blocks as needed. This technique is also known as Application Specific Standard Parts (ASSP). The advantages of this technique are good performance gain of the chip, small footprint, compact IC, and relatively lower NRE cost than a full-custom implementation
  (e.g., between $10 k and 100 k). The disadvantages include still relatively larger time-to-market and NRE (compared to FPGA or COTS microcontrollers).

Microcontroller Unit (MCU or µC)

Newer microcontroller units (MCU) are a “computer on a chip”! This is the most popular types of microcontrollers for complex embedded system design.

Most commonly known microcontrollers are this type.

The MCU contains a microprocessor unit, one or more oscillators (clock, timer), memory (non-volatile program memory and volatile RAM), Analog to digital (A/D) converter, and input/output (I/O) ports.

Some MCU also includes some Digital Signal Processing (DSP) hardware such as digital filters, Fast Fourier Transform (FFT), and digital to analog (D/A) converter.

MCUs have distinct advantages of being very inexpensive and allowing a very short development time (i.e., low NRE and time-to-market). The disadvantages are mainly related low computational resources, lower speed, and lower performance.

Various Microcontroller Unit (MCU or uC)

- ATMEL
- PIC
- MSP
- ARM
- ATOM
- PSoC

Some Popular Microcontrollers

- Intel 8051
- Microchip PIC
- Atmel AVR
- ARM

Various Microcontrollers

Comparison of a few popular microcontrollers – 8051, PIC, AVR, ARM

Main Difference between AVR, ARM, 8051 and PIC Microcontrollers

	8051	PIC	AVR	ARM
Bus width	8-bit for standard core	8/16/32-bit	8/32-bit	32-bit mostly also available in 64-bit
Communication Protocols	UART, USART,SPI,I2C	PIC, UART, USART, LIN, CAN, Ethernet, SPI, I2S	UART, USART, SPI, I2C, (special purpose AVR support CAN, USB, Ethernet)	UART, USART, LIN, I2C, SPI, CAN, USB, Ethernet, I2S, DSP, SAI (serial audio interface), IrDA
Speed	12 Clock/instruction cycle	4 Clock/instruction cycle	1 clock/ instruction cycle	1 clock/ instruction cycle
Memory	ROM, SRAM, FLASH	SRAM, FLASH	Flash, SRAM, EEPROM	Flash, SDRAM, EEPROM
ISA	CLSC	Some feature of RISC	RISC	RISC
Memory Architecture	Harvard architecture	Von Neumann architecture	Modified	Modified Harvard architecture
Power Consumption	Average	Low	Low	Low

[https://www.elprocus.com/]

Various Microcontrollers

Comparison of a few popular microcontrollers – 8051, PIC, AVR, ARM

	8051	PIC	AVR	ARM
Families	8051 variants	PIC16,PIC17, PIC18, PIC24, PIC32	Tiny, Atmega, Xmega, special purpose AVR	ARMv4,5,6,7 and series
Community	Vast	Very Good	Very Good	Vast
Manufacturer	NXP, Atmel, Silicon Labs, Dallas, Cyprus, Infineon, etc.	Microchip Average	Atmel	Apple, Nvidia, Qualcomm, Samsung Electronics, and TI etc.
Cost (as compared to features provide)	Very Low	Average	Average	Low
Other Feature	Known for its Standard	Cheap	Cheap, effective	High speed operation Vast
Popular Microcontrollers	AT89C51, P89v51, etc.	PIC18fXX8, PIC16f88X, PIC32MXX	Atmega8, 16, 32, Arduino Community	LPC2148, ARM Cortex-M0 to ARM Cortex-M7, etc.

Various Microcontrollers

- ATMEL

- Atmel MCU, developed by Atmel Corporation, are excellent for low-power and low-resource embedded systems.

- A popular Atmel MCU model is AVR which provides products (e.g., ATtiny, ATmega) in 8, 16, and 32-bit versions. The IP core is a derivative of Intel 8051 microprocessor architecture.
  - These MCUs supports up to 20 MHz clock rate.
  - Memory range are between 0.5 and 16 kB of Flash.
  - The MCUs typically include I/O ports like Pulse Width Modulation (PWM), Analog to Digital Converter (ADC), Universal Synchronous/Asynchronous Receiver/Transmitter (USART), and Serial Peripheral Interface (SPI).

- Advantage: These MCUs can be very small-in-size (as low as 8-pin packages) and consumes very low power - which are ideal for a lot of ES applications. They are also very inexpensive.
  Disadvantage: These are low-complexity MCU, hence cannot handle high workload or processing requirements.

- Atmel provides a free Integrated Development Environment (IDE) software, called "Atmel Studio," for programming and debugging of these MCUs. Recently, Atmel has been acquired by Microchip Technology.

Various Microcontrollers

- Programmable Interface Controller (PIC)

- A different type of microcontrollers known as Programmable Interface Controller (PIC), developed by Microchip Technology.

- It is an efficient low-power, low-resource, and low-cost option.

- Newer PIC MCU contains 16-bit CPU in Harvard architecture with RISC ISA. Early models used ROM or EPROM for memory, but newer models contain flash memories.

- Most of these MCU have common I/O ports such as I2C, SPI, USB, ADC, PWM, and UART.

- The Integrated Development Environment (IDE) for this MCU is MPLab.

Various Microcontrollers

- MSP

- Texas Instruments (TI) had developed an extremely low-power microcontroller series (MSP 430) that is an attractive option for very low power operation with this highly capable mixed-signal microcontroller.

- It uses Von Neumann Architecture with 16-bit RISC (MIPS-like) ISA. It has 16 registers and uses Little Endian memory. The rich configuration of these series includes a range of flash memory from 1 and 512 kB, ROM options from 1 and 66 kB, RAM options of 128–10 kB, and ADC options of 10, 12, or 14 bit SAR type, as well as 16 or 24 bit Sigma–Delta type. Typical I/O ports include UART, I2C, USB, and even capacitive touch sensing port.

- The Integrated Development Environment (IDE) tool for this series is called Code Composer Studio.

Various Microcontrollers

- Acron RISC Machine (ARM)

- ARM (Acorn RISC Machine) is licensed by ARM Holdings, and manufactured by various chip companies including Atmel, Silicon labs, TI, Kinetis, Freescale, Cypress, STMicroelectronics, Samsung, Renesas Electronics Corporation (which is a spin-off of NEC), etc.

- There are various ARM core designs that ranges from ARMv1 to ARMv8. They are either 16, 32, or 64-bit processors (i.e., widths of the registers).

- As the name suggest, the processor uses ARM ISA. There are both flavors of architecture available, for example ARMv7 is Von Neumann, whereas ARMv9 is Harvard architecture.
- The chips come in three typical configurations:
- Application (A), Real-time (R), and Microcontroller (M).

- Almost 95% of smartphones are using ARM processors.

- The Integrated Development Environment (IDE) for ARM processors is called ARM Development Studio.

Various Microcontrollers

- ATOM
  - Intel Corp. has suffered from its legacy issues (such as backward compatibility, CISC ISA) that complicated its entrance to microcontroller domain.
  - Recently they have developed an ultra-low power MCU with 45 and 22 nm CMOS technologies that has a high processing power. The processor is named as ATOM. This supports 32 and 64-bit hardware, and compatible with ARMv7 or higher MCUs. I/O ports available are SPI, JTAG, SMBus, PCI, etc. They also contain some advanced peripherals such as RTC, WDT, and DDR2 memory.
  - The Integrated Development Environment (IDE) for this MCU is Eclipse.

Various Microcontrollers

- Programmable System on a chip (PSoC)
  - Programmable System on a Chip (PSoC) has been developed by Cypress Semiconductor.
  - These processors are low power and contains reconfigurable analog units that leads to a unique design potential reducing the component counts on the PCB and simplifying the design process. Nonetheless the processors consume higher power compared to competitors (such as MSP 430) due to analog components. Typical current consumption in active mode is ~2 mA, while sleep mode can reduce current to as low as 1–3 µA.
  - There are four models of PSoC: PSoC1 (contains a 8-bit M8C core), PSoC3 (contains a 8-bit 8051 core), PSoC4 (contains a 32-bit ARM Cortex M0 core at 48 MHz), and PSoC5 (contains a 32-bit ARM Cortex M3 that can run at speeds up to 80 MHz). They typically contain ADC, DAC, Analog op-amp, I2C, SPI, USB, UART, and CAN ports.
  - The Integrated Development Environment (IDE) needed for this PSoC1 is called PSoC Designer, while for the rest of the PSoC, the IDE needed is PSoC Creator.

Popular Commercial Prototype Board

|Vahid/Givargis, Embedded Systems Design: A Unified Hardware/Software Introduction|

Popular Commercial Prototype Boards

- Arduino
- Raspberry Pi
- FPGA Boards

Popular Commercial Prototype Boards

- Ardunio
  - Arduino was developed with open source hardware/software concept that has led to its high popularity and low price. It is considered an entry level MCU prototyping board.
  - Popular models are Arduino Uno, Arduino nano, and Arduino mega. They use AVR microcontrollers such as ATmega8, ATmega168, etc.
  - The clock speeds are typically 8 or 16 MHz. The board contains a boot-loader and on-board flash memory of the MCU that makes is simple and easy to use.
  - Also, there are many shield boards (e.g., GPS, Ethernet, LCD, Bluetooth, WiFi) available for most Arduinos that makes rapid prototyping possible.
  - The I/O ports are Analog, GPIO, USB, I2C, SPI, RS-232, PWM, etc. The IDE is called Arduino Sketch.

Popular Commercial Prototype Boards

- Raspberry Pi
  - Raspberry Pi Foundation has developed these boards for promoting educational development and licenses the technology.
  - Common models are: Raspberry Pi 0 (contains ARM MCU with 1 GHz clock and 512 MB RAM), Raspberry Pi 1 A (contains ARMv6 core with 700 MHz clock and 256 or 512 MB RAM), and Raspberry Pi 1 B (contains ARMv7 core with 900 MHz clock and 1 GB RAM).
  - They contain VideoCore IV GPU and USB and audio/video ports.
  - The PIs are programmable with Python and supports C/C++, Java, Perl, and Ruby.

Popular Commercial Prototype Boards

- FPGA
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Selection of MCU

- It is not straightforward to determine the best suited MCU for an application. Many factors must be carefully considered that includes MCU speed, memory size, power consumption, voltage requirement, required ports, etc.
- In some high-speed applications, FPGA or a combination of FPGA with SoC might be suitable.

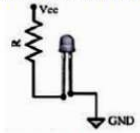
Programming of MCU

- Some MCU prototyping boards, such as Arduino, a simple connection to computer with a USB cable is sufficient. However, some microcontrollers require dedicated programming hardware, such as PICkit 2 for PIC MCU or MiniProg3 for PSoC MCU.
- One advantage of these programming hardware is that it can allow the programmer to debug the content of the MCU registers and variables (called in-circuit debugger), which can be very effective to debug complex code.
- Some boards can contain these hardware within the prototyping board (such as Renesas Prototyping boards). They typically use ICMS or JTAG technique for programming and debugging. The IDE supplied by the vendor can allow access to the in-circuit debugger features.

Sensor

SENSORS & ACTUATORS

- **Sensor**
- A Sensor is used for taking Input
- It is a transducer that converts energy from one form to another for any measurement or control purpose
- Ex. A Temperature sensor
- **Actuator**
- Actuator is used for output.
- It is a transducer that may be either mechanical or electrical which converts signals to corresponding physical actions.
- Ex. LED (Light Emitting Diode)
- LED is a p-n junction diode and contains a **CATHODE** and **ANODE**
- For functioning the anode is connected to +ve end of power supply and cathode is connected to –ve end of power supply.
- The maximum current flowing through the LED is limited by connecting a RESISTOR in series between the power supply and LED as shown in the figure below



- There are two ways to interface an LED to a microprocessor/microcontroller:

1. The Anode of LED is connected to the port pin and cathode to Ground : In this approach the port pin sources the current to the LED when it is at logic high(ie. 1).

2. The Cathode of LED is connected to the port pin and Anode to Vcc : In this approach the port pin sources the current to the LED when it is at logic high (ie. 1). Here the port pin sinks the current and the LED is turned ON when the port pin is at Logic low (ie. 0)

Output and Actuators

In ES systems, there are number of peripherals are commonly found. These can be data storage, output devices, or actuators. Some common examples of these devices are briefly mentioned here.

Output and Actuators

- Data Storage

- Almost all new generation MCU contain onboard memories in the form of
  Random Access Memory (RAM) - Volatile
  Read Only Memory (ROM) – Non-volatile
- External memory like Flash chip, MicroSD card, even Hard Disk Drive (HDD) can be incorporated, if needed. However, it would be best to select an MCU with sufficient onboard memory, if possible.
- Incorporation of MicroSD and HDD will require a file system (such as File Allocation Table, FAT) that makes the overall data storage process slow, complex, and requires overhead. Also, this type of ES with file system will also require operating system (OS).
- On the other hand, advantages of MicroSD and HDD are the possibility of removing from the system for external analysis, very large capacity, and ease of upgrade. The flash chips are non-removable, but fast in operation and lower in power consumption.

Common Memory Types in MCU

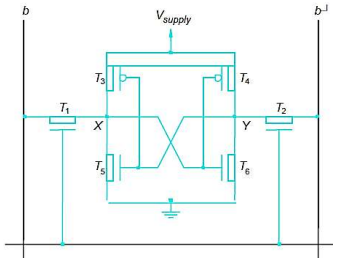
- RAM
  - SRAM
  - DRAM
- ROM
  - MROM
  - PROM
  - EPROM
  - EEPROM
- Flash

Memory

- Random Access Memory (RAM)
  - Volatile or Non-Permanent Memory, i.e., data is lost after the removal of power
  - Can be Written too Many Times
  - It is a general purpose memory which is typically used for storing user data, temporary or Changeable Data
  - Most microcontrollers have some amount of internal RAM. Generally, 1 kbytes ("Embedded" in MCUs) is a common amount, although some microcontrollers have more, some less.

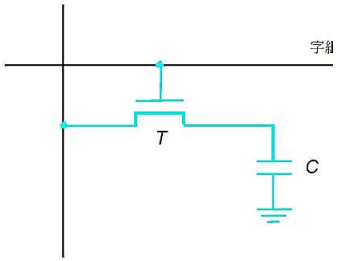
Memory

- SRAM
  - Static Random Access Memory
  - Read/write very fast
  - Needs 6 transistors thus high cost and needs more area
  - Low power consumption
  - Implementation technology
    - CMOS
  - Construct cache memory
  - Access time - 10 nanoseconds



Memory

- DRAM
  - Dynamic Random Access Memory
  - Needs 1 transistor and 1 capacitor
  - Lower cost and compact
  - Each bit must be refreshed periodically
  - Implementation technology
    - CMOS
  - Access time - 50 to 150 nanosecond



- DRAM

❖Dynamic Random Access Memory
❖Needs 1 transistor and 1 capacitor
❖Lower cost and compact
❖Each bit must be refreshed periodically
❖Implementation technology
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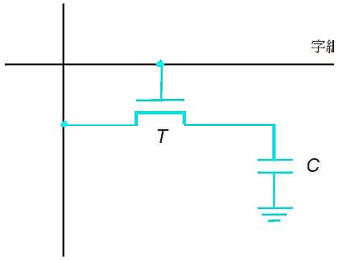


- Read-Only Memory (ROM)

❖ It is non-volatile memory

❖ It is used to store important information which is used to operate the system

❖ As the name refers to read-only memory, we can only read the programs and data stored in it.

❖ It is also called the primary memory unit of the computer system. Also, referred to as permanent memory.

EPROM and EEPROM

- Erasable Programmable Read-Only Memory (EPROM) were the next generation of microcontrollers. It was invented by Dov Frohman of Intel (1971).

- EPROMs were programmable and erasable under ultra-violet (UV) exposure. The programed chip will retain its data even without power (viz. non-volatile).

- Later, another type of reprogrammable ROM became available that could perform the programming operation with a higher electrical voltage (instead of UV light). These chips were known and Electrically Erasable Programmable Read-Only Memory (EEPROM or E2PROM). This are also non-volatile memory by implementing floating-gate transistors (FET) that can be programmed with a higher voltage inducing charge in the gates using field emission technique.

- Newer non-volatile memory chips still using similar techniques (e.g., Flash Memory). Advanced versions of this techniques are still being researched are Ferroelectric RAM (FeRAM) and Magnetoresistive RAM (MRAM).

- Electrically Erasable Programmable Read-Only Memory

❖It is electrically-erasable-and-programmable.

❖Internally, they are similar to **EPROMs**, but the **erase** operation is accomplished **electrically**, rather than by exposure to ultraviolet light. Any byte within an EEPROM may be erased and rewritten.– Erased by higher voltage.

❖The primary tradeoff for this improved

❖functionality is **higher cost**, though **write cycles** are also significantly **longer** than writes to a RAM.

❖EEPROM are typically used to store permanent data.

Memory

- Flash Memory

  - **Flash memory** devices are
    - high density,
    - low cost,
    - nonvolatile,
    - fast (to read, but not to write), and
    - electrically reprogrammable.
    - Write/Erase large blocks of bytes
    - Read bytes
  - EEPROM is similar to **flash memory** (sometimes called flash EEPROM).
  - The principal difference is that
    - EEPROM requires data to be written or erased one byte at a time whereas flash memory allows data to be written or erased in blocks.
  - Flash memory is typically used in MCUs for storing program code

Output and Actuators

- Analog and Digital Actuators

  - Actuators of ES can be either digital or analog types.
    Examples of digital actuators include relays, stepper motors, and buzzers.
    Examples of analog actuators include speakers, and DC motors.
    Actuators can be a combination as well, such as robotic arm drive mechanism.
  - Furthermore, there are a lot of display systems utilized in ES.
    Examples of these are LEDs, seven segment display, LCDs, and graphical display monitors.
  - Some actuators might require special drive circuitry. This is specially true for actuators requiring higher currents than MCU can drive. Typical MCUs (e.g., Arduino) can drive up to 40 mA circuits. If circuits require higher current, driver circuits must be used. Digital drive circuit can be as simple as relay, whereas analog driver circuit might be simple such as power amplifier, or complex such as motor drive circuits. When using actuators that involve magnetic inductor (e.g., relays), ensure to use a back EMF bypassing diode across the actuator.

Output and Actuators

- Output Drive Circuitry and Ports

  - When connecting drive circuitry, note that some MCU output requires external pullup resistors.
  - Carefully study the datasheet of MCU to find out this requirement as well as maximum current ratings.
  - Also pay special attention to available I/O (input/output) ports.
  - These ports can be analog or digital.
    Analog ports are typically connected to ADC or DAC.
    Digital ports can be General Purpose I/O (GPIO), PWM, serial port, parallel port, USB, Ethernet, I2C (or I2C), SPI, SCI, VGA, or COM ports.

Communication Protocols

- Serial Communication
- Parallel Communication

Communication

- Wired Communication Topologies
- Wireless Communication Topologies

Communication

- Wired Communication Topologies

Among wired communication, serial and parallel communications both are used in ES. Common wired communication topologies include

- UART (or USART)
- I2C
- USB
- SCI
- SPI
- Ethernet, etc.

Communication

- Wireless Communication Topologies
  - For newer ES devices, wireless communication techniques are becoming more important as these devices are becoming part of IoT, wearables, smart devices, and sensor net.
  - Wireless technologies are growing in applications. These technologies differ in terms of frequency of operation, communication range, data throughput, and power consumption.
  - Among various wireless communication topologies, RF point-to-point link is not suitable for most ES devices.
  - Some of the more suitable topologies are mentioned below.
    - Cellular Data Network (SIM)
    - WiFi
    - Bluetooth (BT)
    - Sensor Nodes
    - RFID or NFC
    - GPS

Communication

- Wireless Communication Topologies

Performance Metrics

Wireless Security

Peripheral and Bus System

- Peripheral and Bus System
  - Buffer Queue
    - Polling
    - Interrupt
- Direct Memory Access (DMA)
- Timer
- Watchdog Timer (WDT)

Pulse Width Modulation

- Some MCU contains PWM hardware can be utilized for a number of purposes, such as generating various rates of blinking of LEDs by changing frequencies, analog output voltages by filtering the PWM output, and controlling DC motor speed of a RC car by changing the duty cycles of PWM waves.
- Independent PWM channels with programmable period and duty cycles are available in a lot of MCU. Programmable PWM can be enabled or disabled for each channel. Software selection of PWM can also specify duty pulse polarity for each channel.
- Period and duty cycle are double buffered, thus change takes effect when the end of the effective period is reached, i.e., PWM counter reaches to zero or when the channel is disabled.
- Also, programmable center or left-aligned outputs on individual channels can be set. Different bit-width determines PWM resolution or duty cycle ranges. Various MCUs provide a wide range of frequencies for PWM with programmable clock select logic.

Battery Considerations

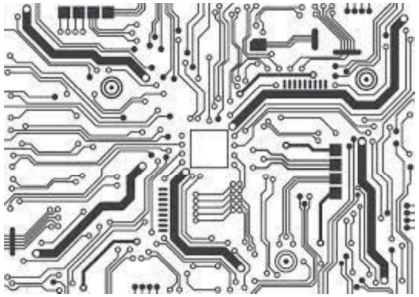
Some ES devices needs to be operated with battery, for example wearables. This section list some of the related considerations that ES designers must be aware of.

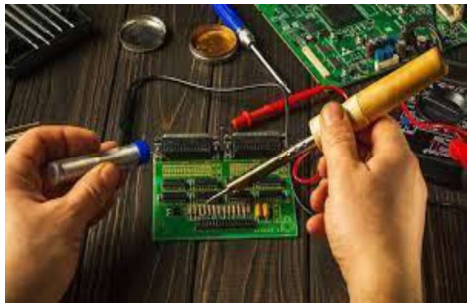
- Battery Topologies
- Power Saving Modes for MCU

Other Hardware

- Power Source
- Clock Oscillator
- Real Time Clock (RTC)
- Reset Circuit, Power-up Reset and watchdog timer Reset
- Memory
- I/O Ports, I/O Buses
- Interrupt Handler
- DAC and ADC
- LCD and LED Display
- Keypad/Keyboard

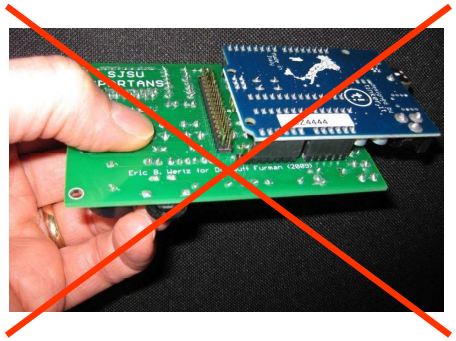
SPICE Simulation and PCB Design

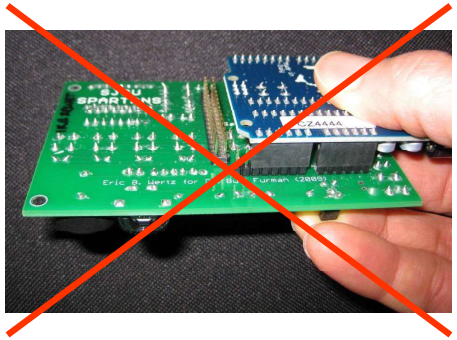




Handling Electronics – How NOT to do it !

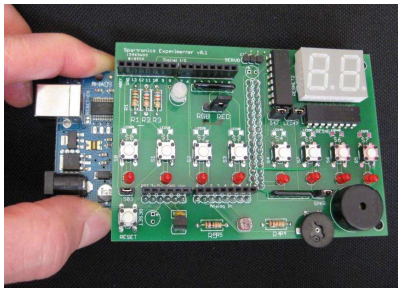
Improper Handling - ***NEVER***!!!

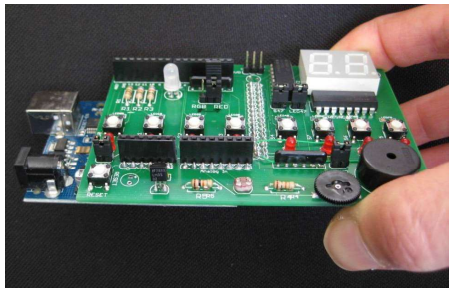




Handling Electronics – The Proper Way

Proper Handling - by the ***edges***!!!





Tools

- Various tools supporting such methodologies
- Commercial and academic
- POLIS (Berkeley), Cierto VCC (Cadence)
- SpecCharts (Irvine)
- STATEMATE, Rhapsody (ilogix)
- Rose RT (Rational)
- SCADE, Esterel Studio (Esterel Technologies)
- Stateflow and Simulink (Mathworks)

Modeling Languages

- Models need to be formal
- Languages for describing models
- Various languages exist
- High level programming languages (C, C++)
- Finite State Machines, Statecharts, SpecCharts, Esterel, Stateflow
- Data Flow Diagrams, Lustre, Signal, Simulink
- Hardware description languages (VHDL, Verilog)
- Unified Modeling Language(UML)

Modeling Languages

- Choice of languages depend upon the nature of computations modeled
- Seq. programming models for standard data processing computations
- Data flow diagrams for iterative data transformation
- State Machines for controllers
- HDLs for hardware components

Website References

Some Website references

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http://www.8051.org
http://www.8051projects.net
http://www.programmersheaven.com
http://microcontrollershop.com

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